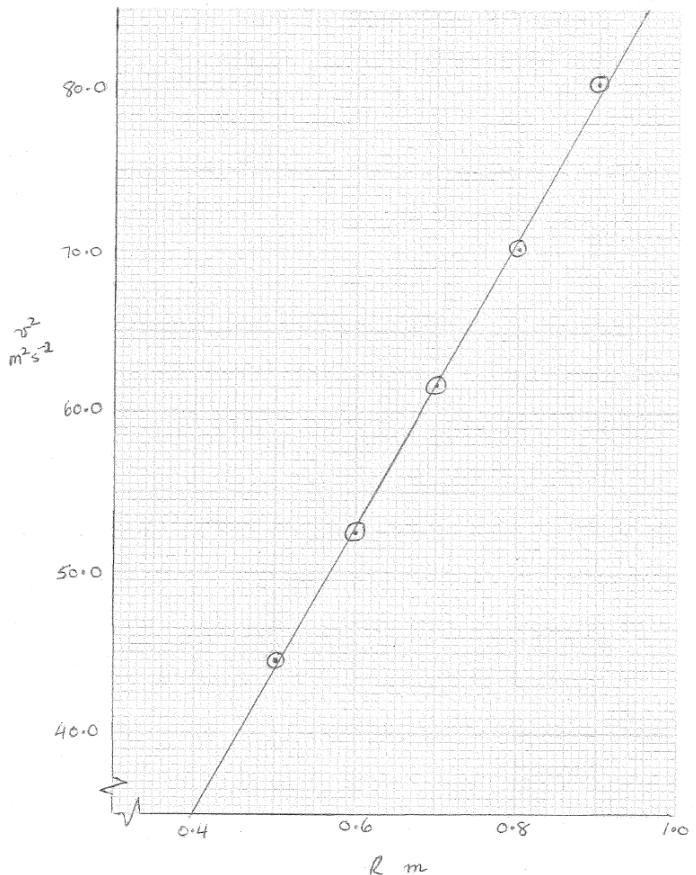


Question			Marking details					Marks available																																			
								AO1	AO2	AO3	Total	Maths	Prac																														
5	(a)		Distance moved by mass P in 1 period, T (i.e. in one rotation) = $2\pi R$ and speed = distance / time = $\frac{2\pi R}{T}$ Alternative: $\omega = \frac{2\pi}{T}$ and $v = \omega R$						1		1	1																															
	(b)		<table><tr><th>R / m</th><th>Time of 10 rot /s</th><th>T / s</th><th>$v / \text{m s}^{-1}$</th><th>$v^2 / \text{m}^2 \text{s}^{-2}$</th></tr><tr><td>0.50</td><td>4.7</td><td>0.47</td><td>6.68</td><td>44.6</td></tr><tr><td>0.60</td><td>5.2</td><td>0.52</td><td>7.25</td><td>52.6</td></tr><tr><td>0.70</td><td>5.6</td><td>0.56</td><td>7.85</td><td>61.6</td></tr><tr><td>0.80</td><td>6.0</td><td>0.60</td><td>8.38</td><td>70.2</td></tr><tr><td>0.90</td><td>6.3</td><td>0.63</td><td>8.98</td><td>80.6</td></tr></table> T column: All values correct (1) For column v all values correct (1) For column v^2 all values correct ecf (1) Consistent use of sig figs in each column and 2 or 3 sig figs (1)					R / m	Time of 10 rot /s	T / s	$v / \text{m s}^{-1}$	$v^2 / \text{m}^2 \text{s}^{-2}$	0.50	4.7	0.47	6.68	44.6	0.60	5.2	0.52	7.25	52.6	0.70	5.6	0.56	7.85	61.6	0.80	6.0	0.60	8.38	70.2	0.90	6.3	0.63	8.98	80.6		4		4	4	4
R / m	Time of 10 rot /s	T / s	$v / \text{m s}^{-1}$	$v^2 / \text{m}^2 \text{s}^{-2}$																																							
0.50	4.7	0.47	6.68	44.6																																							
0.60	5.2	0.52	7.25	52.6																																							
0.70	5.6	0.56	7.85	61.6																																							
0.80	6.0	0.60	8.38	70.2																																							
0.90	6.3	0.63	8.98	80.6																																							

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)	Centripetal force = $0.010 \frac{v^2}{R}$ (1 for $\frac{mv^2}{R}$; 1 if value inserted for m)	1	1		2	1	2
		(ii)	Forces acting on mass Q: $0.090\text{ g} - \tau = 0$ τ : tension (1) So $\tau = 0.090\text{ g}$. Substitution for τ into (c)(i) (1) $0.090\text{ g} = 0.010 \frac{v^2}{R}$ $v^2 = \frac{0.090\text{ g}}{0.010} R$ $v^2 = 9\text{ g } R$ clear and convincing working (1)		3		3	2	3

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
	(d)	(i)	 Axis labels: v^2 $m^2 s^{-2}$ (y-axis), R m (x-axis) Axis scales: Y-axis from 40.0 to 80.0; X-axis from 0.4 to 1.0 Points plotted: (0.45, 45), (0.55, 52), (0.65, 62), (0.75, 70), (0.85, 80) Line of best fit drawn Axes, suitable choice scales (no multiples of 3) and labels on both axes (1) – scales to occupy more than half of paper All points plotted correctly to $\pm\frac{1}{2}$ small square division (2) 4 points plotted correctly to $\pm\frac{1}{2}$ small square division award 1 mark 1-3 points plotted correctly to $\pm\frac{1}{2}$ small square division award 0 marks Line of best fit (1)						
					4		4	4	4

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
		(ii)	gradient = $\frac{50.0}{0.56} = 89.286 \text{ m s}^{-2}$; find gradient from best fit line (1) also gradient = $9g$ general method (1) $9g = 89.286$ $g = \frac{1}{9} 89.286 = 9.92 \text{ m s}^{-2}$ unit mark (1) (Accept $g = 8.8$ to 10.8 m s^{-2} i.e. uncertainty of ~10%) Use of single data point award a maximum of 2 marks			3	3	3	3
		(iii)	Take measurements for each value of R several times or measure time of more rotations or use of video capture or increase radius and period Accept repeat readings Don't accept have an assistant			1	1		1
			Question 5 total	1	13	4	18	15	18